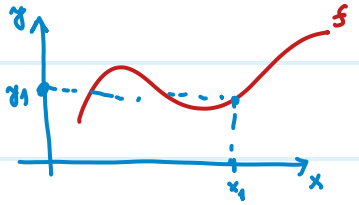


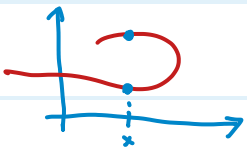
Funkcie viacerých premenných

Funkcia 1 premennej



$y = f(x)$

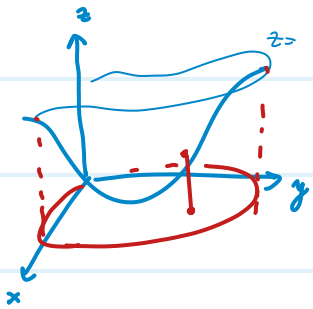
$\forall x \in D(f) \quad \exists! \, y \in H(f)$



toto nie je funkcia

2 premenné

$z = f(x,y)$



$D(f) = \{ [x,y] \in R \times R, \; z = f(x,y) \}$

viac premenných

$y = f(x_1, x_2, x_3, \ldots x_n)$

$x_1, x_2, x_3, \ldots x_n$ – definičný obor

y – funkcia

$z = \ln(x \cdot y)$

$z = \sqrt{x^2+y^2}$

$z = 3$

$z = x + 2$

Príklad.

$$f(x,y) = \frac{\sqrt{x^2+y^2-9}}{x}$$

$D(f) = ?$

nájdi definičný obor

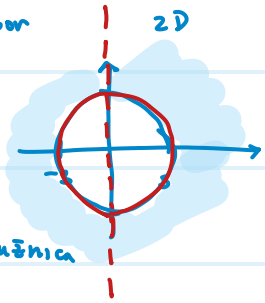
2D

$x \neq 0$

$x^2+y^2-9 \geq 0$

$x^2+y^2 \geq 9$

$x^2+y^2 = 9$ – kružnica



$$f(x,y,z) = \frac{x}{\sqrt{9-x^2-y^2-z^2}}$$

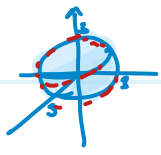
$D(f) = ?$

$9-x^2-y^2-z^2 > 0$

$x^2+y^2+z^2 < 9$

guľa $S = [0,0,0]$

$r = 3$



PRAVIDLÁ. 1. zachováva sa dimenzia

$f+g$ $\quad f \cdot g$ $\quad f^{-1}$ $\quad f \circ g = f(g)$

$f-g$ $\quad \frac{f}{g}$

GRAF funkcie viac premenných **2D**

① GEOGEBRA 3D

② priamka
rovina
kvadratické plochy – kužeľ
hyperboloid
paraboloid
guľa
elipsoid
valcové plochy
+ hyperbolický
eliptický
paraboloid

Contour Map – Vrstevnice

pr $f(x,y)=\sqrt{64-x^2-y^2}$

$63=\sqrt{64-x^2-y^2}$

$63^2-64=-x^2-y^2$

$-\text{číslo}=x^2+y^2$

$[x,y]=[0,0]$

$x^2+y^2=64$

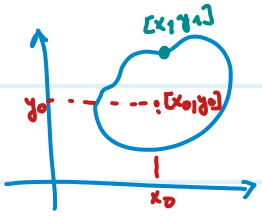






2D

Body v definičnom obore



$f: z=f(x,y)$

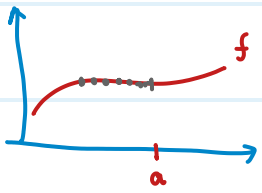
$[x_0,y_0] \in D(f)$
vnútorný bod → interior point

$[x_1,y_1] \in D(f)$
hraničný bod → boundary point
bod na hranici

$[x_2,y_2] \notin D(f)$
vonkajší bod

Limita

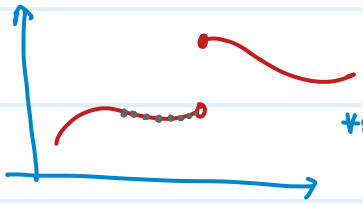
1D



$$\lim_{x \to a} f(x) = \underset{L}{f(a)}$$

$$\{x_i\} \to a$$

$$f(x_1), f(x_2), \ldots f(x_n), \ldots \to f(a)$$



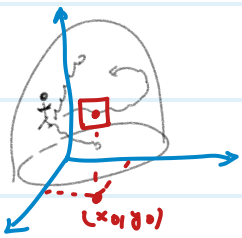
Cauchy

$$\forall \varepsilon > 0 \quad \exists \delta > 0$$

$$|f(x) - L| < \varepsilon$$

$$\lim_{x \to x_0} f(x) = L$$

2D



$$\lim_{(x,y) \to (x_0,y_0)} f(x,y) = L$$

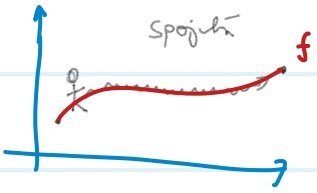
Cauchy

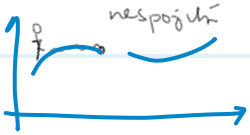
$$\forall \varepsilon > 0 \quad \exists \delta > 0 \text{ takové}$$

$$|f(x,y) - L| < \varepsilon$$

$$0 < \underbrace{f(x,y)}_{\text{tu to musí patřit definiční obor}} < \delta$$

SPOJITOST





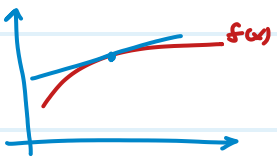
Příklad.

$$\lim_{(x,y) \to (1,2)} \frac{5x^2y}{x^2+y^2} = \frac{5 \cdot 1^2 \cdot 2}{1^2+2^2} = \frac{10}{5} = 2$$

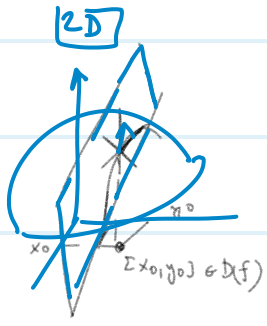
Derivácie

1D

$$y = f(x) \qquad f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$



2D



$$z = f(x,y) \qquad \text{PARCIÁLNE DERIVÁCIE} \quad f'_x, \; f'_y$$

$$\frac{\partial}{\partial x} f(x,y) = \frac{\partial f}{\partial x} = f'_x = \frac{dz}{dx}$$

$$\frac{\partial}{\partial y} f(x,y) = \frac{\partial f}{\partial y} = f'_y = \frac{dz}{dy}$$

PARCIÁLNA DERIVÁCIA V BODE

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = f'_x(x_0,y_0)$$

DEFINÍCIA. $z = f(x,y)$

$$f'_x(x,y) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

$$f'_y(x,y) = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

PRÍKLAD. $f(x,y) = x \cdot e^{x^2+y} + \sin y$

f'_x $\quad f'_x(1,2)$ $\quad f'_y$ $\quad f'_y(2,3)$

R. $\frac{\partial f}{\partial x} = 1 \cdot e^{x^2+y} + x \cdot e^{x^2+y} \cdot 2x + 0$

$x \cdot e^{x^2+2} + \sin 2$

$$\left.\frac{\partial f}{\partial x}\right|_{(1,2)} = 1 \cdot e^{1^2+2} + 1 \cdot e^{1^2+2} \cdot 2 \cdot 1 + 0 =$$

$\frac{\partial f}{\partial y} = x \cdot e^{x^2+y} \cdot (0+1) + \cos y$

$\frac{\partial f}{\partial y} = x \cdot e^{x^2+y} \cdot 1 + \cos y$

$\left.\frac{\partial f}{\partial y}\right|_{(2,3)} = 2 \cdot e^{2^2+3} \cdot 1 + \cos 3$

$\frac{x}{2} \cdot e^{x^2+y} + \sin y$

$\frac{\partial^2 f}{\partial x \partial y}$ $\quad \frac{\partial^2 f}{\partial x \partial x}$